Fpt University

TRƯỜNG ĐẠI HỌC FPT

ISP392 PROJECT REPORT

Software Requirement Specification

HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM

(HSWRMS)

A Wellness Resort & Spa Management System

- Hanoi, Sep 2026 -

Table of Contents

Table of Contents **2**
Change Log **4**
Definition and Acronyms **5**
I. Overall Requirements **5**
1. Context Diagram 5
2. Main Business Flow 6
2.1 Customer Onboarding & Wellness Profile Setup (BF-01) 6
Link Customer Onboarding & Wellness Profile Setup (BF-01) 7
2.2 Retreat Booking, Deposit & Check-In (BF-02) 8
2.3 Spa/Therapy Scheduling & Session Fulfillment (BF-03) 8
2.4 Personalized Dining & F&B Service Fulfillment (BF-04) 10
2.5 Consolidated Checkout, Payment & Customer Feedback (BF-05) 10
2.6 System Administration, Permission & Management Monitoring (BF-06) 11
3. Conceptual Data Model 12
3.1 Entity Relationship Diagram 12
3.2 Entity Descriptions 12
3.3 Data Business Rules 12
4. User Requirements 12
4.1 Actors 12
4.2 Use Cases (UC) 13
4.3 Use Case Diagrams 15
4.3.1 Module 1 - Authentication & Sensitive Health Profile: (Sai) 15
4.3.2 Module 2 - Retreat Package & Accommodation Booking: 17
4.3.3 Module 3 - Spa & Therapy Scheduling: 18
4.3.4 Module 4 - Dietary F&B Management: 19
4.3.5 Module 5 - Consolidated Checkout & Analysis: 20
4.3.6 Module 6 - System Administration & Permission Management: 21
4.3.7 Consolidated System-Wide Use Case View 21
4.4 Permission Matrix 22
5. System Functionalities 22
5.1 Screen Inventory 22
5.2 External API Inventory 22
5.3 Background Job Inventory 22
II. Functional Requirements **22**
1. Module 1 - Authentication & Sensitive Health Profile 22
1.1 UC01 - Register, Verify Email & Login 22
1.2 UC02 - Manage Dietary & Health Profile 22
1.3 UC03 - Manage Staff Accounts & Roles 22
1.4 UC04 - Manage Master Data 22
1.5 UC05 - Request Health Data Deletion 22

2. Module 2 - Retreat Package & Accommodation Booking....22
2.1 UC06 - Browse Retreat Packages.... 22
2.2 UC07 - Book Retreat & Pay Deposit.... 22
2.3 UC08 - Check-In & Assign Villa....22
2.4 UC09 - Manage Villa Status.... 22
2.5 UC10 - View Booking & Itinerary....22
3. Module 3 - Spa & Therapy Scheduling....22
3.1 UC11 - Schedule Included Spa/Therapy Session.... 22
3.2 UC12 - Match Therapist & Treatment Room.... 22
3.3 UC13 - View Therapist Schedule & Health Notes.... 22
3.4 UC14 - Mark Session Completed / No-Show.... 22
3.5 UC15 - Manually Book Extra Spa Service....22
4. Module 4 - Dietary F&B Management.... 22
4.1 UC16 - Pre-select Daily Meals.... 22
4.2 UC17 - View Daily Meal Prep Dashboard.... 23
4.3 UC18 - Update Meal Order Status.... 23
4.4 UC19 - Order Extra A-la-carte Item....23
4.5 UC20 - Minimize Health Data for Chef....23
5. Module 5 - Consolidated Checkout & Analytics....23
5.1 UC21 - Generate Consolidated Invoice....23
5.2 UC22 - Process Final Payment & Update Villa.... 23
5.3 UC23 - Submit Review & Rating.... 23
5.4 UC24 - View Revenue Dashboard.... 23
5.5 UC25 - Export Occupancy & Therapist Utilization....23
6. Module 6 - System Administration & Permission Management....23
6.1 UC26 - Configure Business Parameters.... 23
6.2 UC27 - Toggle Feature Flags.... 23
6.3 UC28 - Configure Permission Matrix....23
6.4 UC29 - Delegate Limited Admin Rights....23
6.5 UC30 - View Sensitive-Data Audit Log....23
6.6 UC31 - View System Overview Dashboard....23
7. Background Job Requirements.... 23
8. External API Requirements.... 23
III. Requirement Appendix.... 23
1. Business Rules....23
2. System Messages.... 23
3. Other Requirements.... 23
4. Open Items Requiring Team Confirmation.... 23
5. References and Source Basis.... 23

Change Log

Date	A/M/D	In charge	Change Description
Sep 8, 2026	A	Tất Đạt	Created SRS's structure baseline from the project specification and current six-module plan.
Sep 12, 2026	M	Tất Đạt	Renamed project to HeavenStay Wellness Resort Management System (HSWRMS).
Sep 12, 2026	A	Phương Thảo	Added BF04_Personalized Dining & F&B Service Fulfillment(2.4)
Sep 12, 2026	A	Tất Đạt	Added BF03_Spa/Therapy Scheduling & Session Fulfillment(2.3)
Sep 13, 2026	A	Xuân Hòa	Added BF05_Consolidated Checkout, Payment & Customer Feedback(2.5)
Sep 13, 2026	A	Đức Hoàng	Added BF01_Customer Onboarding & Wellness Profile Setup(2.1)
Sep 14, 2026	A	Gia Đạt	Added BF02_Retreat Booking, Deposit & Check-In(2.2)
Sep 14, 2026	A	Thành Vinh	Added BF06_Customer Onboarding & Wellness Profile Setup(2.6)
Sep 15, 2026	A	Tất Đạt	Added HSWRMS Context Diagram(1.)
Sep 15, 2026	A	Tất Đạt	Added M3 Spa Scheduling Use Case Diagram(4.3.3)
Sep 15, 2026	A	Xuân Hòa	Added Module 5 - Consolidated Checkout & Analysis:(4.3.5)
Sep 15, 2026	A	Phương Thảo	Add Module 4- Dietary F&B Management (4.3.4)
Sep 15, 2026	A	Đức Hoàng	Add Module 1 - Authentication & Sensitive Health Profile (4.3.1)
Sep 16, 2026	A	Gia Đạt	Add Module 2 - Retreat Package & Accommodation Booking (4.3.2)
Sep 16, 2026	A	Thành Vinh	Add Module 6 - System Administration & Permission Management (4.3.6)

A = Added M = Modified D = Deleted

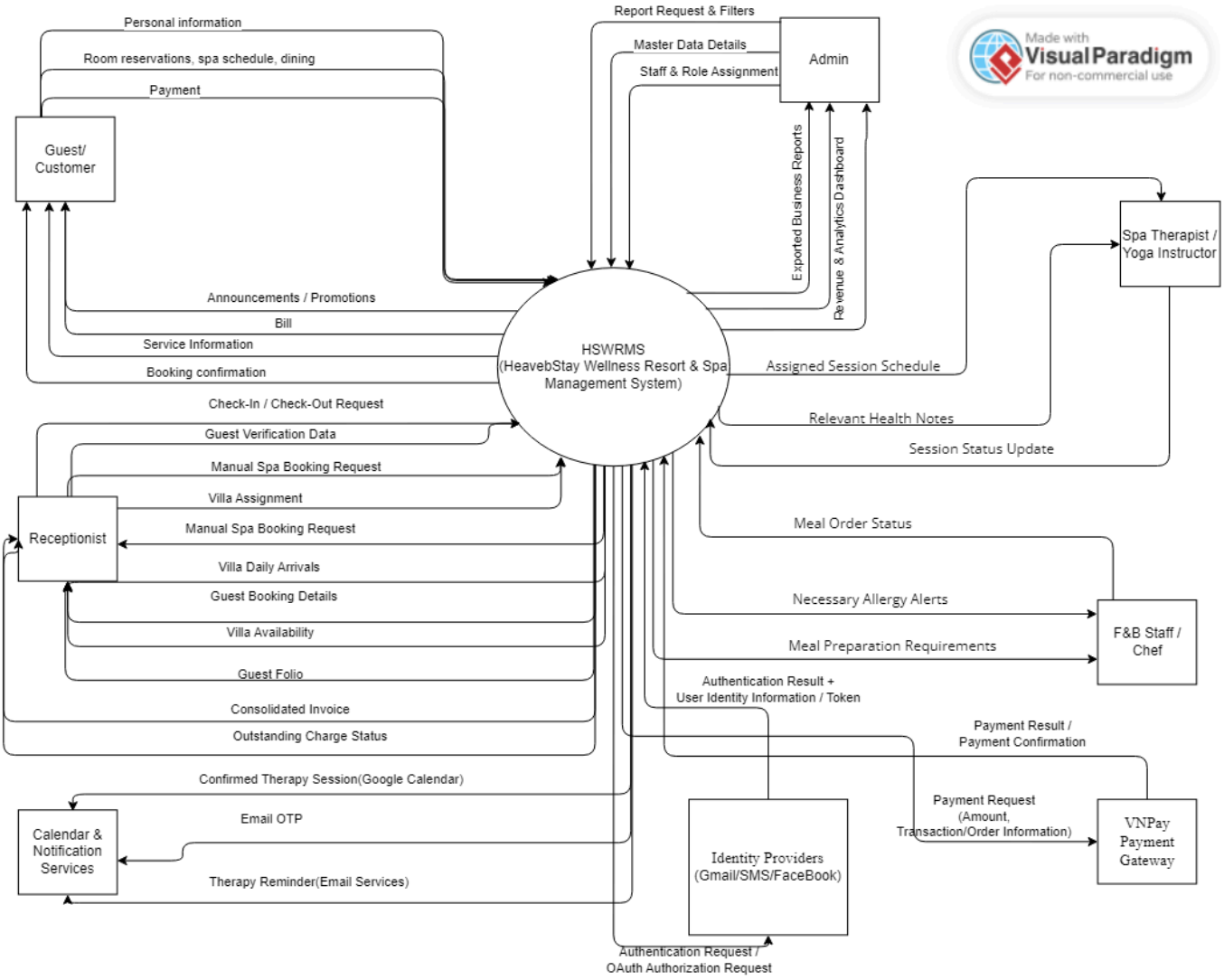
Definition and Acronyms

Acronym / Term	Definition
AWRMS	Aurevia Wellness Resort Management System
SRS	Software Requirement Specification
PMS	Property Management System
RBAC	Role-Based Access Control
OTP	One-Time Password
SSO	Single Sign-On
F&B	Food & Beverage
API	Application Programming Interface
ERD	Entity Relationship Diagram
CRUD	Create, Read, Update, Delete
PII	Personally Identifiable Information
GWI	Global Wellness Institute
AHLEI	American Hotel & Lodging Educational Institute
UC	Use Case
BF	Business Flow
2D Lock	Atomic reservation of both an available Therapist and an available Treatment Room for one requested slot
Folio	Guest central account aggregating package, spa and F&B charges for checkout
VNPay	Confirmed payment gateway for deposit and final checkout payment

I. Overall Requirements

1. Context Diagram

HeavenStay Wellness Resort Management System (HSWRMS) is an integrated platform for wellness retreat booking, accommodation operations, spa/therapy scheduling, personalized nutrition, consolidated billing, analytics, and permission-controlled administration. It connects guests and resort staff with external payment, calendar, email, and identity services.



HSWRMS_Context_diagram

Figure 1. HSWRMS system context diagram

System boundary rationale. Guest, Receptionist, Therapist, F&B Staff/Chef, Admin and Manager are external human actors because they initiate or receive business interactions with AWRMS. VNPay, Calendar & Notification Services, and Google/Facebook Identity Providers remain outside the system boundary because AWRMS consumes their services rather than implementing those external platforms.

2. Main Business Flow

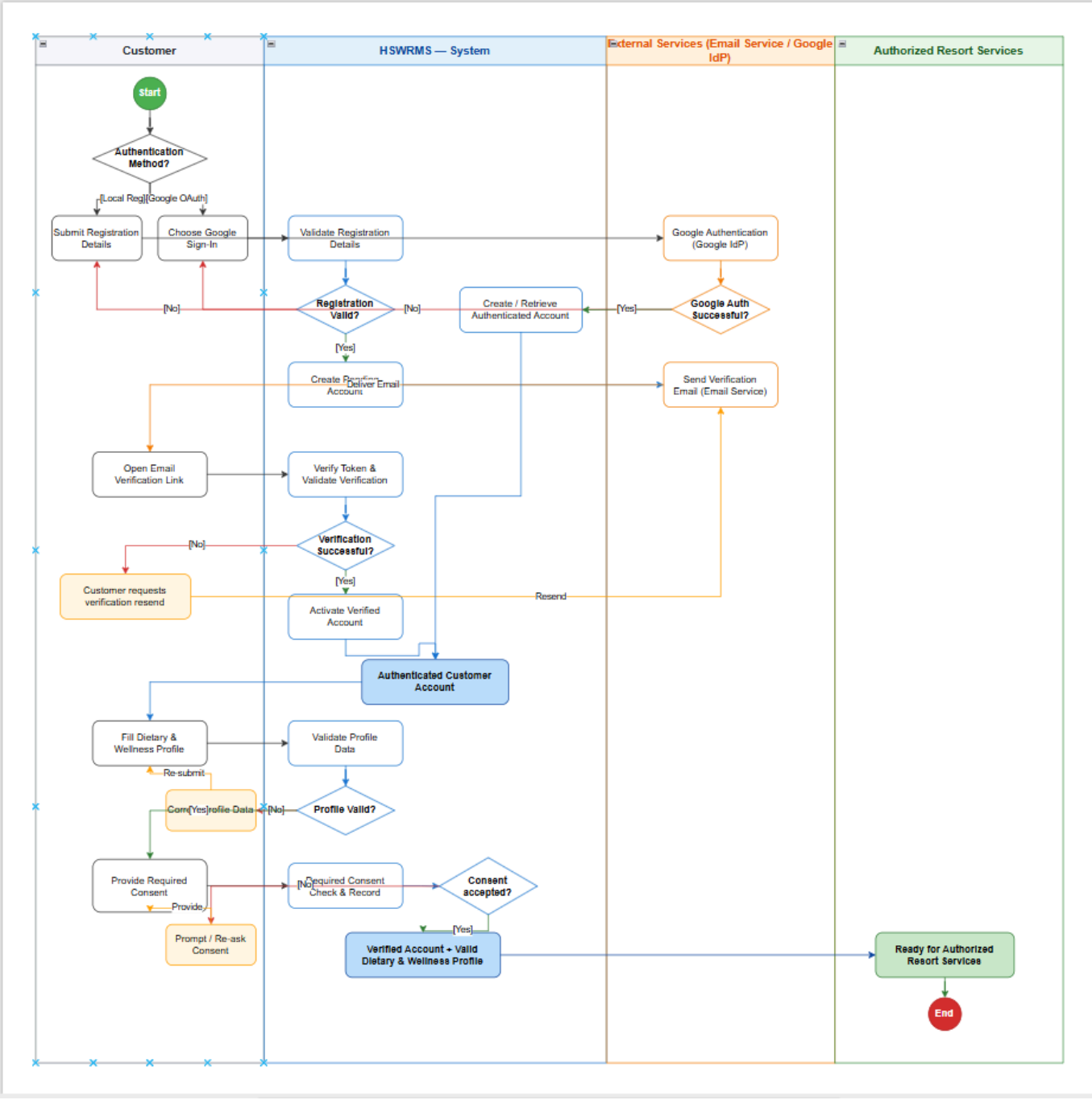
2.1 Customer Onboarding & Wellness Profile Setup (BF-01)

Primary Module: M1 - Authentication & Sensitive Health Profile

Primary Actors / Systems: Customer, Admin, HSWRMS, Google Identity Providers, Email Notification Service

Trigger: A new Customer wants to create an account and use HSWRMS.

End Condition: The Customer has a verified account and a valid Dietary & Health Profile that can be securely used by authorized resort services.



Link Customer Onboarding & Wellness Profile Setup (BF-01)

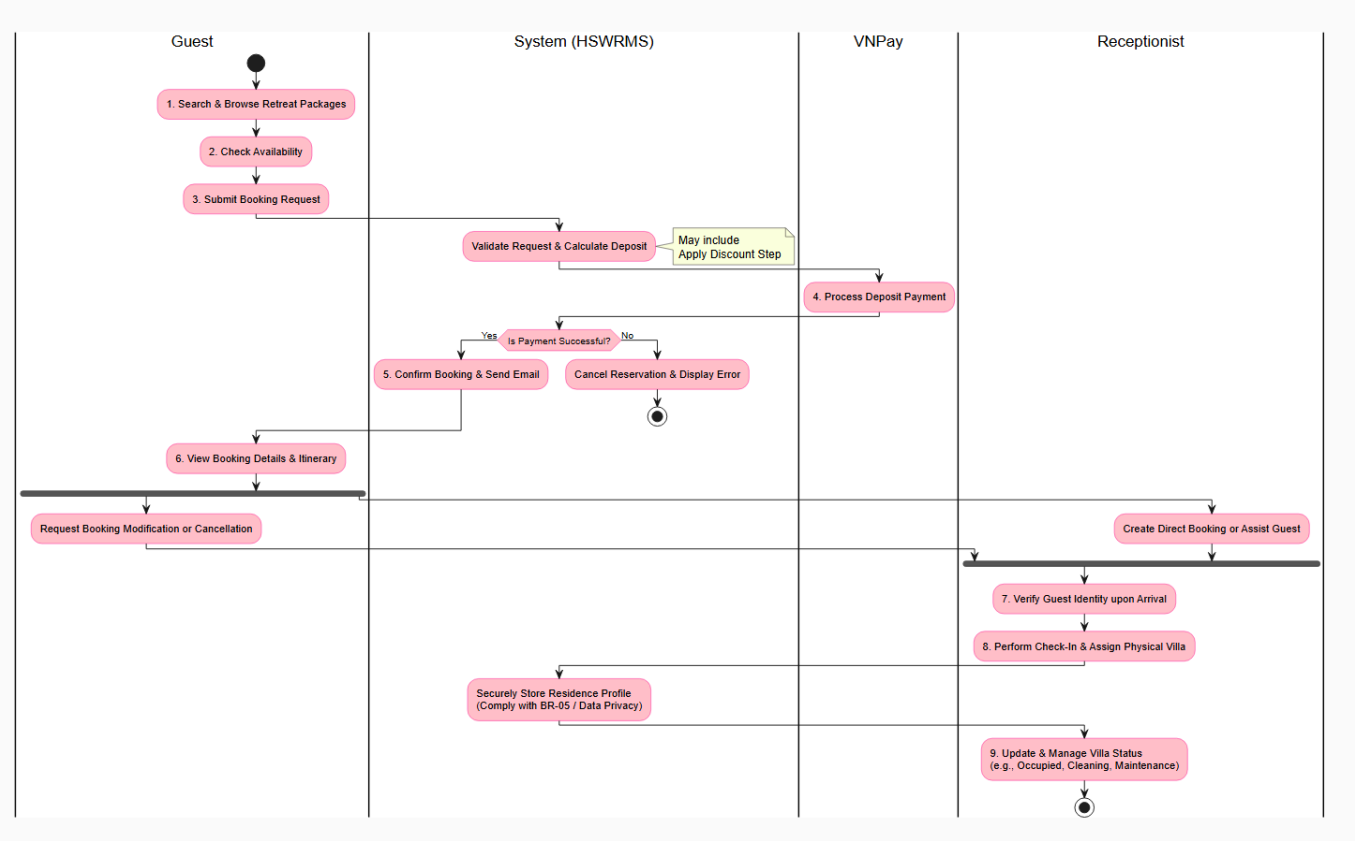
2.2 Retreat Booking, Deposit & Check-In (BF-02)

Primary Module: M2 - Retreat Package & Accommodation Booking

Primary Actors / Systems: Customer, Receptionist, HSWRMS, VNPay Payment Gateway

Trigger: An authenticated Customer wants to book a wellness retreat.

End Condition: The retreat booking is confirmed, required deposit is successfully paid, and the Customer is checked in with an assigned physical villa.



2.3 Spa/Therapy Scheduling & Session Fulfillment (BF-03)

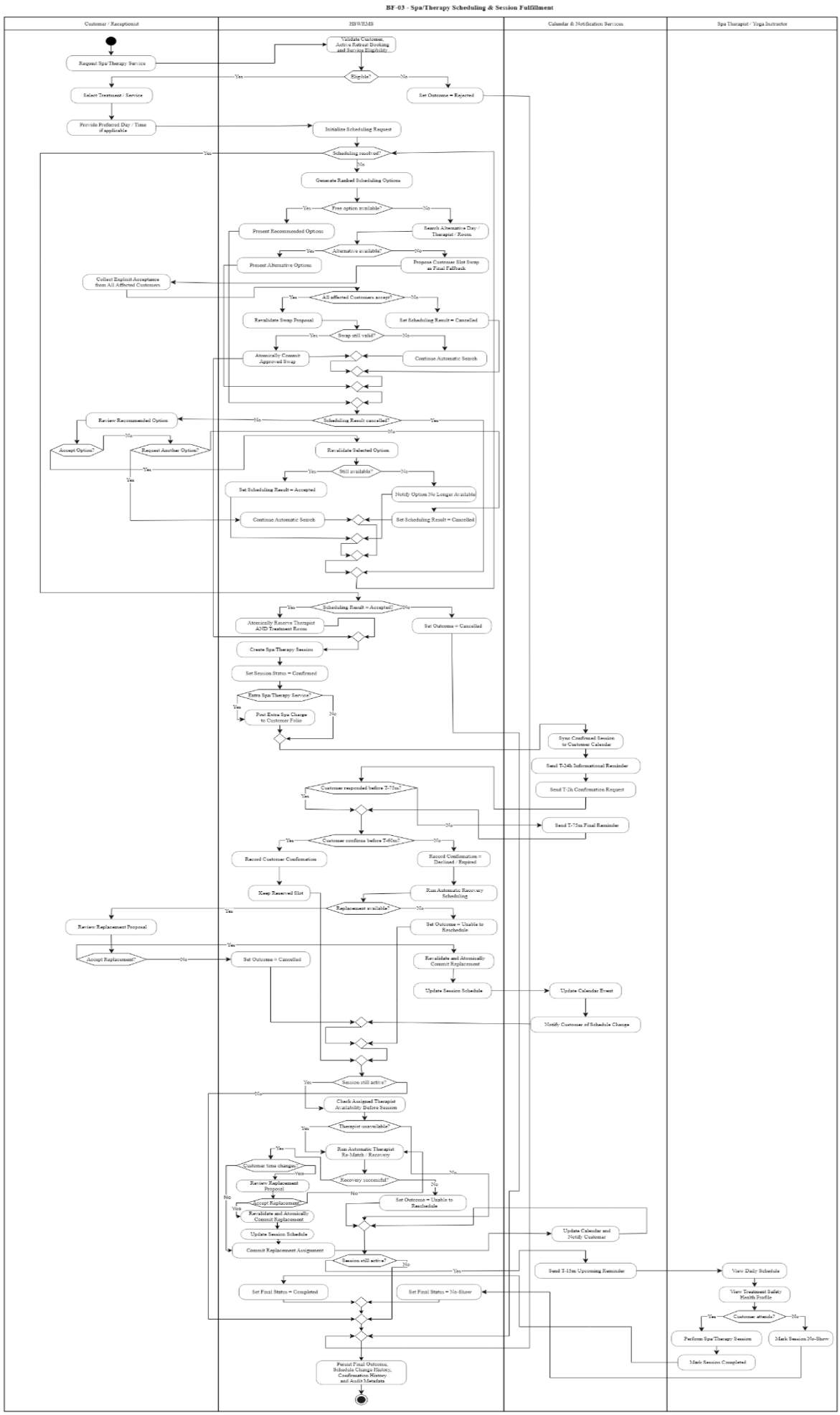
Primary Module: M3 - Spa & Therapy Scheduling (CORE)

Primary Actors / Systems: Customer, Receptionist, Spa Therapist / Yoga Instructor, HSWRMS, Calendar & Notification Services

Trigger: An authenticated Customer, or a Receptionist acting on behalf of an eligible Customer, requests a Spa/Therapy service.

Flow Summary: _HSWRMS validates Customer and service eligibility, then automatically generates suitable scheduling options combining a time slot, qualified Therapist/Yoga Instructor, and compatible Treatment Room. The Customer/Receptionist reviews and confirms the final option. Before confirmation, HSWRMS revalidates availability and reserves the Therapist and Treatment Room atomically. If no suitable option exists, the system automatically searches alternatives and may use Customer slot swapping as a final fallback with explicit consent from all affected Customers.
_For confirmed sessions, HSWRMS synchronizes calendar information and manages pre-session reminders and confirmation. A declined/expired confirmation or Therapist unavailability triggers automatic recovery and rescheduling. Any accepted schedule change is updated and communicated to the Customer.
_Before fulfillment, the Therapist/Yoga Instructor views the assigned schedule and required Treatment Safety Health Profile, performs the session, and records the final attendance outcome as Completed or No-Show.

End Condition: BF-03 ends through a single final path after HSWRMS records the final outcome and relevant scheduling/confirmation history. The outcome may be **Rejected**, **Cancelled**, **Unable to Reschedule**, **Completed**, or **No-Show**.


BF-03 - Spa/Therapy Scheduling & Session Fulfillment
Customer / Receptionist
HSWR24S
Calendar & Notification Services
Spa Therapist / Yoga Instructor
Request Spa/Therapy Service
Validate Customer, Active Retreat Booking and Service Eligibility
Eligible?
Yes
No
Select Treatment / Service
Set Outcome = Rejected
Provide Preferred Day / Time if applicable
Initialize Scheduling Request
Scheduling resolved?
Generate Ranked Scheduling Options
Free option available?
Present Recommended Options
Search Alternative Day / Therapist / Room
Alternative available?
Present Alternative Options
Propose Customer Slot Swap as Final Fallback
Collect Explicit Acceptance from All Affected Customers
All affected Customers accept?
Revalidate Swap Proposal
Set Scheduling Result = Cancelled
Swap still valid?
Atomically Commit Approved Swap
Continue Automatic Search
Review Recommended Option
Scheduling Result cancelled?
Accept Option?
Request Another Option?
Revalidate Selected Option
Still available?
Set Scheduling Result = Accepted
Notify Option No Longer Available
Continue Automatic Search
Set Scheduling Result = Cancelled
Scheduling Result = Accepted?
Atomically Reserve Therapist AND Treatment Room
Set Outcome = Cancelled
Create Spa/Therapy Session
Set Session Status = Confirmed
Extra Spa/Therapy Service?
Post Extra Spa Charge to Customer Folio
Sync Confirmed Session to Customer Calendar
Send T-24h Informational Reminder
Send T-2h Confirmation Request
Customer responded before T-75m?
Send T-75m Final Reminder
Customer confirms before T-60m?
Record Customer Confirmation
Record Confirmation = Declined / Expired
Keep Reserved Slot
Run Automatic Recovery Scheduling
Replacement available?
Review Replacement Proposal
Set Outcome = Unable to Reschedule
Accept Replacement?
Set Outcome = Cancelled
Revalidate and Atomically Commit Replacement
Update Session Schedule
Update Calendar Event
Notify Customer of Schedule Change
Session still active?
Check Assigned Therapist Availability Before Session
Therapist unavailable?
Run Automatic Therapist Re-Match / Recovery
Customer time changes?
Recovery successful?
Review Replacement Proposal
Accept Replacement?
Set Outcome = Unable to Reschedule
Revalidate and Atomically Commit Replacement
Update Session Schedule
Commit Replacement Assignment
Update Calendar and Notify Customer
Session still active?
Send T-15m Upcoming Reminder
View Daily Schedule
View Treatment Safety Health Profile
Customer attends?
Perform Spa/Therapy Session
Mark Session No-Show
Mark Session Completed
Set Final Status = Completed
Set Final Status = No-Show
Persist Final Outcome, Schedule Change History, Confirmation History and Audit Metadata

Link Spa/Therapy Scheduling & Session Fulfillment (BF-03)

2.4 Personalized Dining & F&B Service Fulfillment (BF-04)

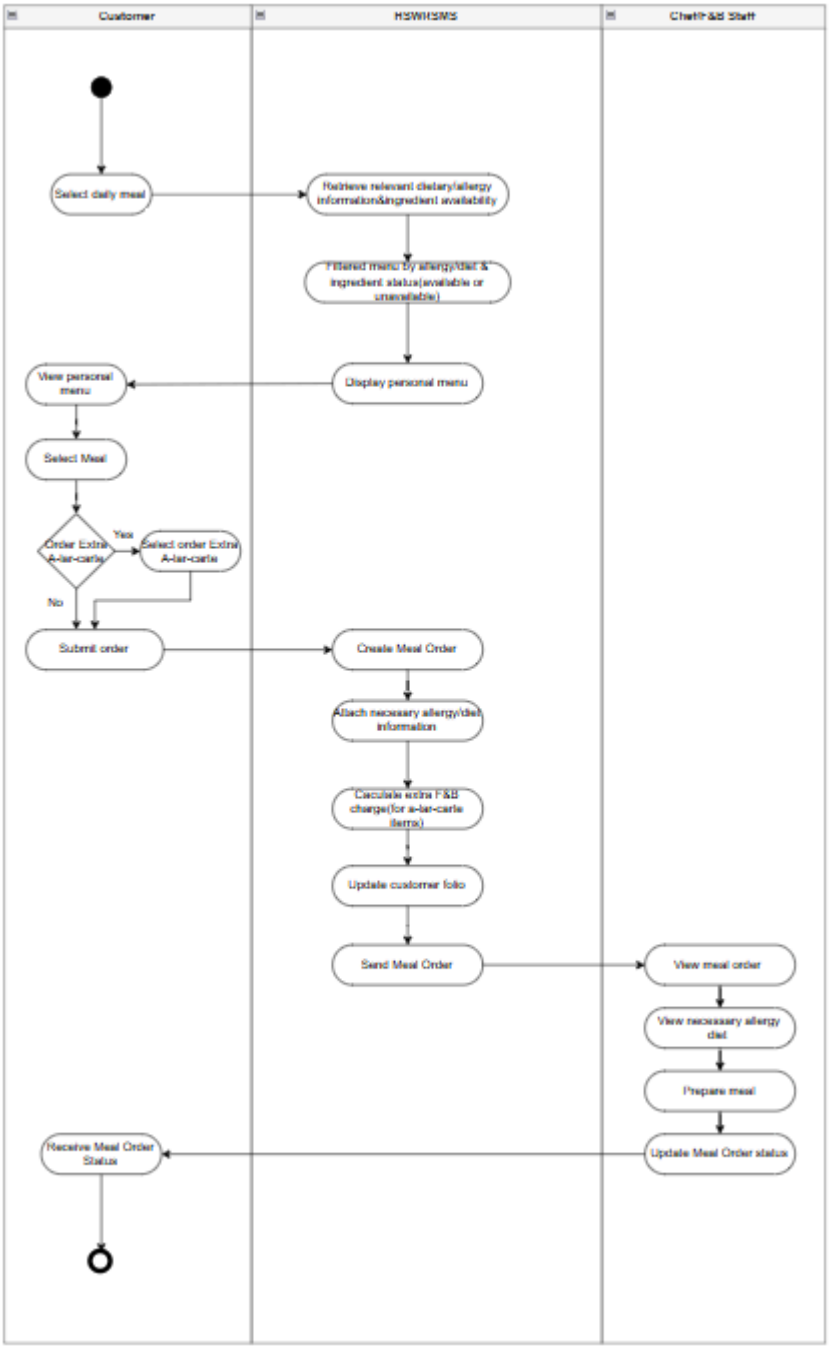
Primary Module: M4 - Dietary F&B Management

Primary Actors / Systems: Customer, Chef / F&B Staff, HSWRMS(System)

Precondition: F&B Staff/Chef periodically checks ingredient quantity and expiry date and updates ingredient availability in the system. The system uses the latest availability information to determine whether menu items are Available or Unavailable.

Trigger: A Customer wants to select or order a meal during the retreat.

End Condition: The selected meal is prepared and fulfilled, the final meal order status is updated, and any applicable Extra A-la-carte charge is recorded in the Customer Folio.



Link Personalized Dining & F&B Service Fulfillment

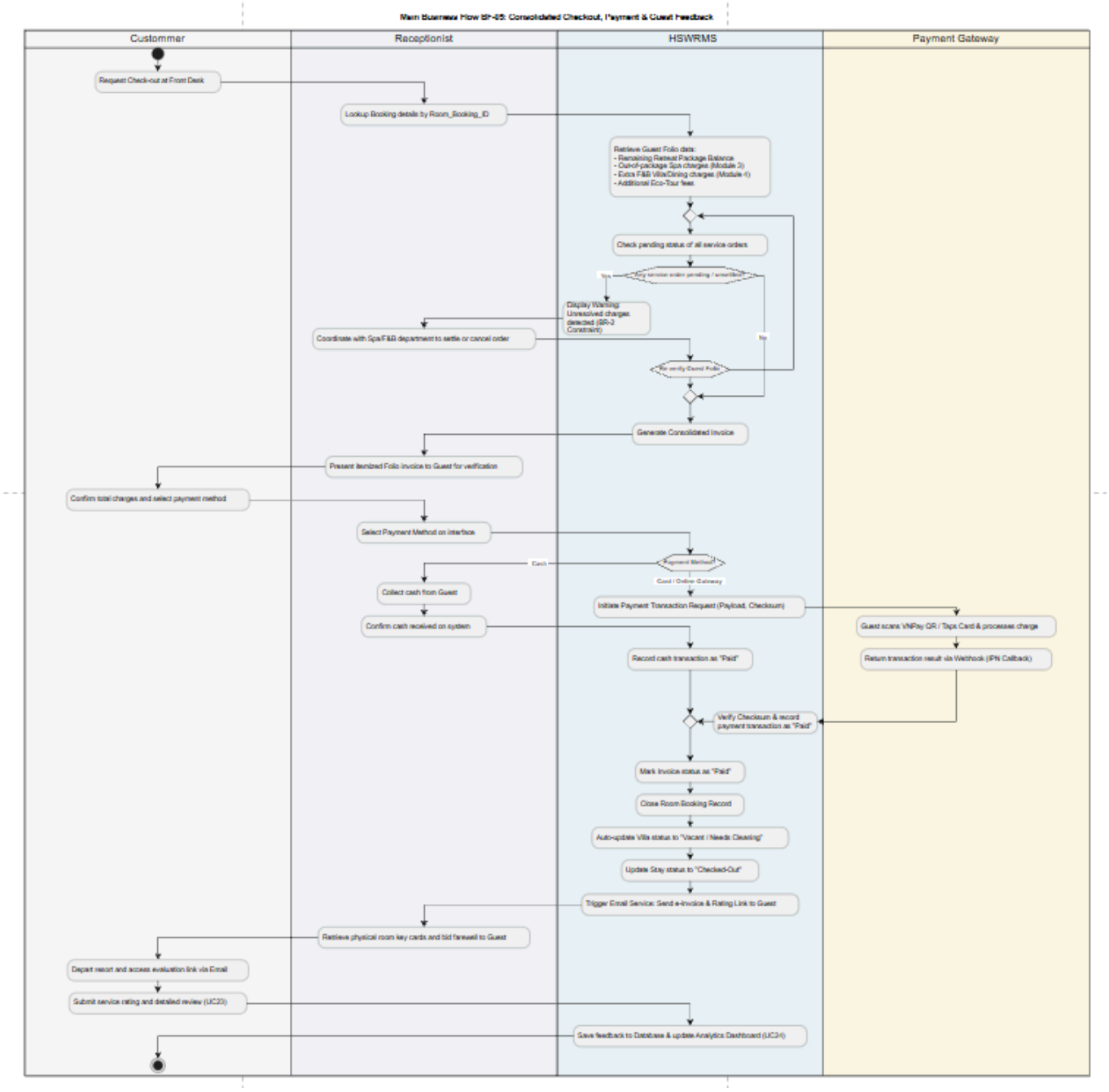
2.5 Consolidated Checkout, Payment & Customer Feedback (BF-05)

Primary Module: M5 - Consolidated Checkout & Analytics

Primary Actors / Systems: Custmer, Receptionisot, HSWRMS, VNPay Payment Gateway

Trigger: Customer requests to check out from the resort.

End Condition: All outstanding charges are settled, checkout is completed, villa status is updated, and the Customer may submit a review/rating.



Link Consolidated Checkout, Payment & Customer Feedback (BF-05)

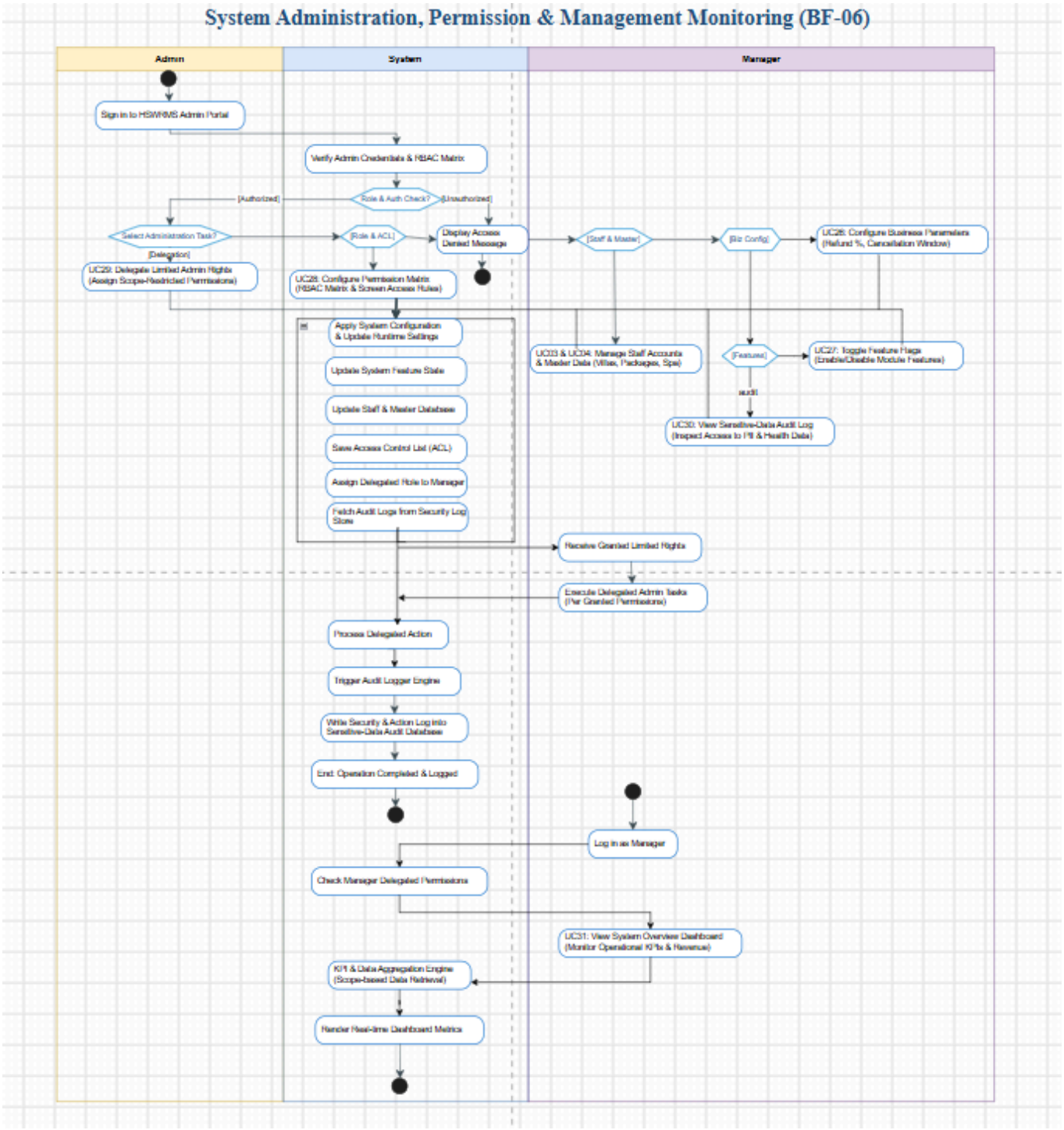
2.6 System Administration, Permission & Management Monitoring (BF-06)

Primary Module: M6 - System Administration & Permission Management

Primary Actors / Systems: Admin, Manager, HSWRMS

Trigger: Admin or Manager signs in to perform an authorized administrative or management operation.

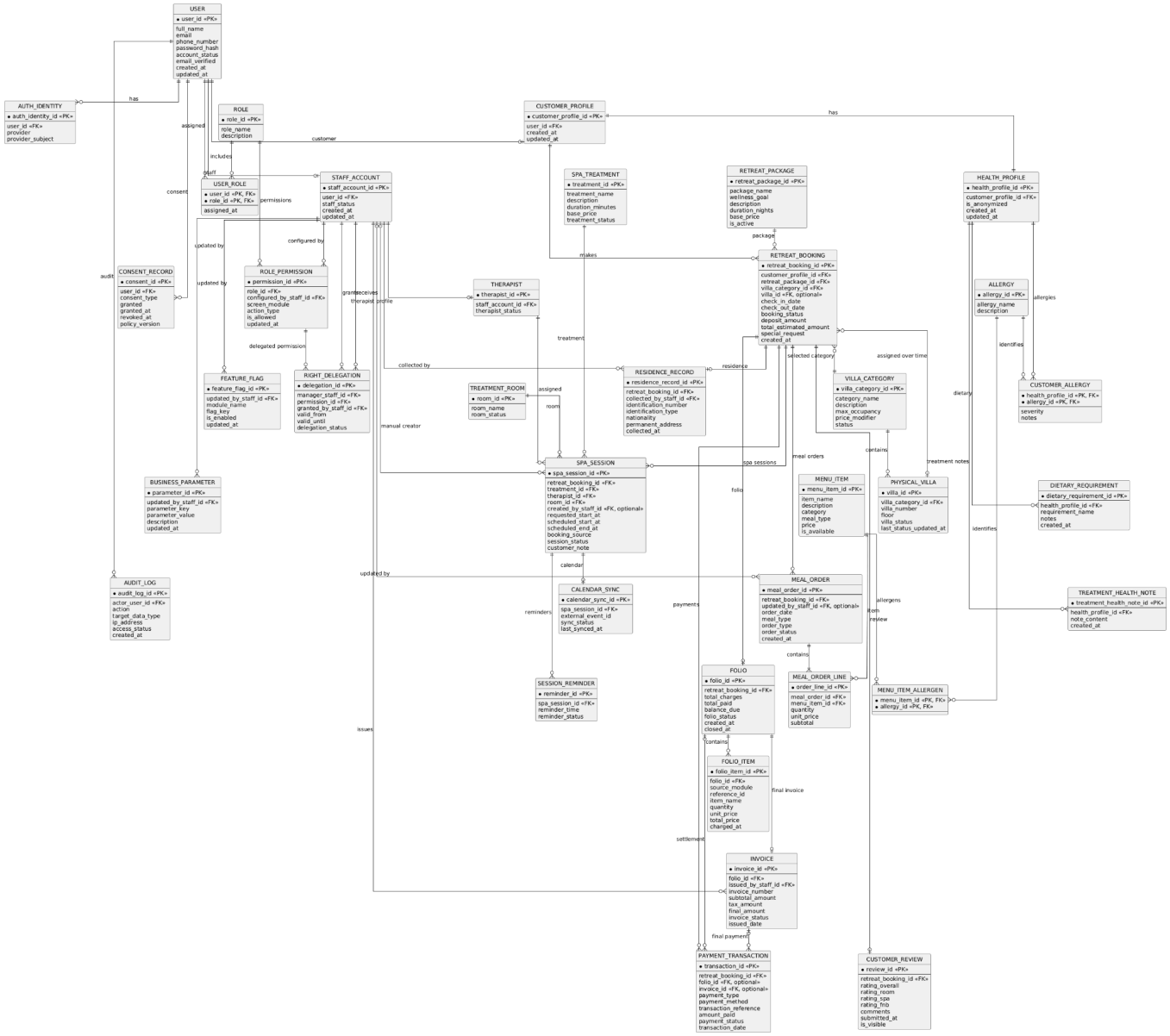
End Condition: Authorized configuration changes are applied and audited, or requested management information is displayed/exported according to the user's permission level.



Link System Administration, Permission & Management Monitoring (BF-06)

3. Conceptual Data Model

3.1 Entity Relationship Diagram



[Link Logical ERD]

3.2 Entity Descriptions

1. Entities: User

Attribute	Value
Purpose	Stores user accounts used to authenticate and access the HSWRMS.
Key business attributes	Full name, email, phone number, password, account status, and email verification status.
Business identity	A unique user account identified by `user_id`.
Status / lifecycle	An account can be created, verified, activated, deactivated, or locked depending on its account status.

2. Entities:Auth-identity

Attribute	Value
Purpose	Stores external authentication identities linked to a user account, such as Google or Facebook.
Key business attributes	`auth_identity_id`, (`user_id`), provider, and provider subject.
Business identity	A unique external identity identified by `auth_identity_id` and the `provider + provider_subject` combination.
Status / lifecycle	Created upon linking provider; removed/disabled on unlinking.

3. Entities: Role

Attribute	Value
Purpose	Defines roles used to control user access and permissions in HSWRMS.
Key business attributes	`role_id`, role name, and role description.
Business identity	A unique role identified by `role_id` and `role_name`.
Status / lifecycle	Created when an external identity is linked to a user account and remains associated until the identity is unlinked.

4. Entities: User_Role

Attribute	Value
Purpose	Associates users with roles for role-based access control.
Key business attributes	`user_id`, `role_id`, and assignment date.
Business identity	A unique user-role assignment identified by the combination of `user_id` and `role_id`.
Status / lifecycle	Created when a role is assigned to a user and remains valid while the assignment is maintained.

5. Entities: Customer_Profile

Attribute	Value
Purpose	Stores business-related profile information associated with a Customer account.
Key business attributes	`customer_profile_id`, (`user_id`), creation date, and update date.
Business identity	A unique Customer profile identified by `customer_profile_id` and linked to a `user_id`.
Status / lifecycle	Created when a Customer profile is established and updated throughout the Customer relationship.

6. Entities:Staff_Account

Attribute	Value
Purpose	Stores staff-specific account information used by operational and administrative staff.
Key business attributes	`staff_account_id`, (`user_id`), staff status, creation date, and update date.
Business identity	A unique staff account identified by `staff_account_id` and linked to a `user_id`.
Status / lifecycle	Created → active/inactive/suspended according to staff status.

7. Entities: Health_Profile

Attribute	Value
Purpose	Stores health-related information associated with a Customer for wellness and service personalization.
Key business attributes	`health_profile_id`,(`customer_profile_id`), anonymization status, creation date, and update date.
Business identity	A unique health profile identified by `health_profile_id` and associated with a Customer profile.
Status / lifecycle	Created → maintained during the Customer relationship → anonymized when applicable.

8. (*)Entities: Consent_Record X

Attribute	Value
Purpose	Records Customer consent related to the collection and processing of personal or health data.
Key business attributes	`consent_id`, (`user_id`), consent type, consent status, granted date, revoked date, and policy version.
Business identity	A unique consent record identified by `consent_id`.
Status / lifecycle	Created with granted/revoked state and updated when consent is changed or withdrawn.

9. Entities:Allergy X

Attribute	Value
Purpose	Maintains the master list of allergies used by the health and F&B modules.
Key business attributes	`allergy_id`, allergy name, and description.
Business identity	A unique allergy identified by `allergy_id` and allergy name.
Status / lifecycle	Defined and maintained as master data.

10. Entities: Customer_Allergy X

Attribute	Value
Purpose	Records allergies associated with a Customer's health profile.
Key business attributes	`health_profile_id`, `allergy_id`, severity, and notes.
Business identity	A unique Customer-allergy association identified by the combination of `health_profile_id` and `allergy_id`
Status / lifecycle	Created when an allergy is recorded and maintained while the allergy information remains applicable.

11. Entities: Dietary_Requirement

Attribute	Value
Purpose	Stores dietary requirements associated with a Customer's health profile for personalized F&B services.
Key business attributes	`dietary_requirement_id`,(`health_profile_id`), requirement name, notes, and creation date.
Business identity	A unique dietary requirement identified by `dietary_requirement_id`.
Status / lifecycle	Created → maintained → updated when the Customer's dietary requirements change.

12. Entities: Treatment_Health_Note X

Attribute	Value
Purpose	Stores health notes relevant to Spa and Therapy services without exposing the Customer's full health profile.
Key business attributes	`treatment_health_note_id`,(`health_profile_id`), note content, and creation date.
Business identity	A unique treatment health note identified by `treatment_health_note_id`.
Status / lifecycle	Created and maintained for treatment-related use.

13. Entities: Retreat package

Attribute	Value
Purpose	Defines wellness retreat packages offered by the resort.
Key business attributes	`retreat_package_id`, package name, wellness goal, description, duration, base price, and active status.
Business identity	A unique retreat package identified by `retreat_package_id` and package name.
Status / lifecycle	Defined → active/inactive depending on availability for booking.

14. Entities: Villa_Category

Attribute	Value
Purpose	Defines categories of villas available for retreat accommodation.
Key business attributes	`villa_category_id`, category name, description, maximum occupancy, price modifier, and status.
Business identity	A unique villa category identified by `villa_category_id` and category name.
Status / lifecycle	Maintained as active/inactive accommodation master data.

15. Entities: Physical_Villa

Attribute	Value
Purpose	Represents individual physical villas available within the resort.
Key business attributes	`villa_id`, (`villa_category_id`), villa number, floor, villa status, and last status update time.
Business identity	A unique physical villa identified by `villa_id` and villa number.
Status / lifecycle	Available → occupied/maintenance → available or inactive depending on villa status.

16. Entities: Retreat_booking

Attribute	Value
Purpose	Records a Customer's retreat booking and provides the stay context referenced by accommodation, Spa, F&B, and billing processes.
Key business attributes	`retreat_booking_id`, check-in date, check-out date, booking status, deposit amount, estimated total amount, and special request.
Business identity	A unique retreat booking identified by `retreat_booking_id`.
Status / lifecycle	Pending → confirmed → checked-in → completed, with cancellation possible according to booking rules.

17. Entities: Residence_Record

Attribute	Value
Purpose	Records Customer identification and residence information collected during the retreat stay process.
Key business attributes	`residence_record_id`, identification number, identification type, nationality, permanent address, and collection time.
Business identity	A unique residence record identified by `residence_record_id`.
Status / lifecycle	Created when residence information is collected and retained according to applicable data-management rules.

18. Entities: Therapist

Attribute	Value
Purpose	Represents staff members qualified to provide Spa or Therapy services.
Key business attributes	`therapist_id` and therapist status.
Business identity	A unique therapist identified by `therapist_id`.
Status / lifecycle	Active/inactive/suspended according to therapist status.

19. Entities: Spa_Treatment

Attribute	Value
Purpose	Defines Spa and Therapy services offered by the resort.
Key business attributes	`treatment_id`, treatment name, description, duration, base price, and treatment status.
Business identity	A unique treatment identified by `treatment_id` and treatment name.
Status / lifecycle	Active/native according to treatment availability

20. Entities: Treatment_Room

Attribute	Value
Purpose	Represents physical rooms used for Spa and Therapy sessions.
Key business attributes	`room_id`, room name, and room status.
Business identity	A unique treatment room identified by `room_id` and room name.
Status / lifecycle	Available → occupied/maintenance → available or inactive.

21. Entities: Spa_Session

Attribute	Value
Purpose	Records scheduled Spa or Therapy sessions for Customers during their retreat stay.
Key business attributes	`spa_session_id`, scheduled start time, scheduled end time, booking source, session status, and Customer note.
Business identity	A unique Spa session identified by `spa_session_id`.
Status / lifecycle	Requested → confirmed/scheduled → completed, with possible cancellation or no-show states.

22. Entities: Calendar_Syns

Attribute	Value
Purpose	Stores synchronization information between confirmed Spa sessions and an external calendar service.
Key business attributes	`calendar_sync_id`, external event ID, synchronization status, and last synchronization time.
Business identity	A unique calendar synchronization record identified by `calendar_sync_id`.
Status / lifecycle	Pending → synced or failed; may be cancelled when the related session is cancelled.

23. Entities: Session_Reminder

Attribute	Value
Purpose	Stores reminder information for scheduled Spa or Therapy sessions.
Key business attributes	`reminder_id`, reminder time, and reminder status.
Business identity	A unique session reminder identified by `reminder_id`.
Status / lifecycle	Pending → sent or failed; may be cancelled when the related session is cancelled.

24. Entities: Menu_Item

Attribute	Value
Purpose	Stores food and beverage items available through the resort's F&B services.
Key business attributes	`menu_item_id`, item name, description, category, meal type, price, and availability status.
Business identity	A unique menu item identified by `menu_item_id` and item name.
Status / lifecycle	Available/unavailable depending on menu availability.

25. Entities: Menu_Item_Allergen

Attribute	Value
Purpose	Associates menu items with allergens to support allergy-aware food selection.
Key business attributes	`menu_item_id`, `allergy_id`.
Business identity	A unique menu-item/allergen association identified by the combination of `menu_item_id` and `allergy_id`.
Status / lifecycle	Created and maintained as menu allergen information changes.

26. Entities: Meal_Order

Attribute	Value
Purpose	Records Customer F&B orders associated with a retreat stay, including daily meals and extra F&B orders.
Key business attributes	`meal_order_id`, order date, meal type, order type, order status, and creation date.
Business identity	A unique meal order identified by `meal_order_id`.
Status / lifecycle	Pending → preparing → ready → delivered, with possible cancellation.

27. Entities: Meal_Order_Line

Attribute	Value
Purpose	Stores individual menu items included in an F&B order.
Key business attributes	`order_line_id`, quantity, unit price, and subtotal.
Business identity	A unique order line identified by `order_line_id`.
Status / lifecycle	Follows the lifecycle of its associated meal order.

28. Entities: Folio

Attribute	Value
Purpose	Consolidates charges generated during a Customer's retreat stay for checkout and billing.
Key business attributes	`folio_id`, total charges, total paid, balance due, folio status, creation date, and closing date.
Business identity	A unique folio identified by `folio_id`.
Status / lifecycle	Open → partially paid/paid → closed.

29. Entities: Folio_Item

Attribute	Value
Purpose	Records individual charges include in Customer’s Folio
Key business attributes	`folio_item_id`, source module, reference ID, item name, quantity, unit price, total price, and charge date.
Business identity	A unique folio item identified by `folio_item_id`.
Status / lifecycle	Created when a charge is added to the folio and retained as part of the billing history.

30. Entities: Invoice

Attribute	Value
Purpose	Represents the consolidated invoice generated from a Customer's folio.
Key business attributes	`invoice_id`, invoice number, subtotal, tax amount, final amount, invoice status, and issue date.
Business identity	A unique invoice identified by `invoice_id` and invoice number.
Status / lifecycle	Draft → issued → paid or cancelled.

31. Entities: Payment_Transaction

Attribute	Value
Purpose	Records payments related to retreat bookings, folios, or invoices.
Key business attributes	`transaction_id`, payment type, payment method, transaction reference, amount paid, payment status, and transaction date.
Business identity	A unique payment transaction identified by `transaction_id` and transaction reference where applicable.
Status / lifecycle	Pending → successful or failed; successful transactions may later be refunded or cancelled according to payment rules.

32. Entities:Customer_Review

Attribute	Value
Purpose	Stores Customer feedback and ratings submitted after completing a retreat stay.
Key business attributes	`review_id`, overall rating, room rating, Spa rating, F&B rating, comments, submission date, and visibility status.
Business identity	A unique review identified by `review_id`.
Status / lifecycle	Created after Customer submission → visible/hidden according to review visibility.

33. Entities: Role_Permission

Attribute	Value
Purpose	Defines permissions that control which actions a role can perform within system modules and screens.
Key business attributes	`permission_id`, module/screen, action type, permission status, and update date.
Business identity	A unique permission configuration identified by `permission_id`.
Status / lifecycle	Created → configured/updated as part of RBAC administration.

34. Entities: Feature_Flag

Attribute	Value
Purpose	Controls the activation or deactivation of configurable system features.
Key business attributes	`feature_flag_id`, module name, flag key, enabled status, and update date.
Business identity	A unique feature flag identified by `feature_flag_id` and `flag_key`.
Status / lifecycle	Enabled/disabled and updated as system configuration changes.

35. Entities: Business_Parameter

Attribute	Value
Purpose	Stores configurable business parameters used to control system behavior.
Key business attributes	`parameter_id`, parameter key, parameter value, description, and update date.
Business identity	A unique business parameter identified by `parameter_id` and `parameter_key`.
Status / lifecycle	Created → updated as business configuration changes.

36. Entities: Right_Delegation

Attribute	Value
Purpose	Records temporary delegation of permissions to authorized management staff.
Key business attributes	`delegation_id`, valid-from date, valid-until date, and delegation status.
Business identity	A unique delegation identified by `delegation_id`.
Status / lifecycle	Pending/active → expired or revoked according to the delegation period and status.

37. Entities: Audit_Log

Attribute	Value
Purpose	Records system access and administrative activities for auditing and traceability.
Key business attributes	`audit_log_id`, action, target data type, IP address, access status, and creation time.
Business identity	A unique audit record identified by `audit_log_id`.
Status / lifecycle	Created when an auditable action or access event occurs and retained as historical audit data.

3.3 Data Business Rules

Rule	Description	Enforced On
1. Unique account email		

4. User Requirements

4.1 Actors

#	Actor	Description
0	Guest	Browses the public landing page, retreat packages, accommodation, spa, dining, and other promotional information.May start registration, sign-in, password recovery, or supported SSO authentication, but cannot access authenticated Customer functions until successfully authenticated.
1	Customer	An authenticated resort customer who manages account, health and dietary information; browses and books retreat packages; views booking and itinerary information; requests and confirms Spa/Therapy sessions; selects meals and orders additional F&B items; reviews folio and payment information; completes payment; and may submit a post-checkout review.
2	Receptionist	Handles front-desk and stay operations, including daily arrivals, Customer verification, check-in/check-out, required residence information, villa assignment and villa status. The Receptionist may initiate or confirm Spa/Therapy bookings on behalf of a Customer and book extra Spa services, while therapist/room matching and scheduling remain system-driven. The Receptionist also supports folio and consolidated checkout operations.
3	Spa Therapist / Yoga Instructor	Views assigned daily Spa/Therapy sessions and the Customer's Treatment Safety Health Profile required to identify contraindications and safely perform the assigned treatment. Unrelated administrative, residence, payment, and account-security information remains hidden.The Therapist/Yoga Instructor performs the assigned session and records the final attendance outcome as Completed or No-Show.

#	Actor	Description
4	F&B Staff / Chef	Manages meal-preparation operations;views aggregated meal requirements and Customer allergy/diet information required for food preparation; monitors ingredient/menu availability where permitted; prepares personalized meals and updates meal/order preparation status. Unrelated medical, identity, residence, or payment information is not exposed.
5	Admin	Performs system-administration and governance functions, including staff-account administration, supported master data, business parameters, feature flags, permission configuration, limited administrative-right delegation, and sensitive-data audit monitoring. Admin capabilities remain subject to mandatory privacy, security, and business rules.
6	Manager	Uses management dashboards, reports, and approved system-overview functions to monitor revenue, occupancy, therapist workload, and other aggregated operational information. Manager access is primarily read-only or restricted and does not include staff-account, role, or permission administration by default. Specific limited administrative capabilities may be delegated by Admin when explicitly authorized.
7	VNPay Payment Gateway	External payment service used to process supported retreat deposit and final-payment transactions and return payment results to HSWRMS. A successful payment transaction does not by itself complete the resort checkout process.
8	Calendar & Notification Services	External supporting services used by HSWRMS to synchronize confirmed Spa/Therapy sessions with Customer calendar events and deliver reminders, confirmation requests, OTPs, and schedule-change notifications where applicable. HSWRMS remains the source of truth for scheduling decisions and session status.
9	Identity Providers	External identity services such as Google and Facebook that authenticate a Visitor through supported SSO flows and return an authentication result to HSWRMS. After successful authentication and account mapping, the user accesses the system as a Customer.

4.2 Use Cases (UC)

ID	Use Case	Feature	Use Case Description
	Register Account	M1 Authentication & Health Profile	Guest creates a new account by providing the required registration information.
	Verify Email	M1	Guest verifies the registered email address using the verification process supported by the Email Service.
	Forgot / Reset Password	M1	Registered user requests a password reset and establishes a new password through the supported recovery process. .
	Login (Email / Password)	M1	Users authenticate using their registered email address and password to access authorized system functions. .
	Login with Google	M1	Users authenticate through Google Identity Provider and access their authorized system functions using Google SSO. .
	Login with Facebook	M1	Users authenticate through Facebook Identity Provider and access their authorized system functions using Facebook SSO.

ID	Use Case	Feature	Use Case Description
	Manage Account Settings	M1	Customer updates personal information, contact preferences, and account security settings through the Account Settings page.
	Manage Health & Dietary Profile	M1	Customer records and maintains dietary, allergy, and physical treatment information relevant to the retreat experience
	Manage Consent	M1	Customer reviews, provides, and withdraws consent for the collection and use of personal, health, dietary, and allergy information.
	Request Health Data Deletion	M1	Customer requests deletion or anonymization of health and allergy data after retreat completion
	Browse Retreat Packages	M2 Booking & Accommodation	Guest browses packages and filters by wellness goal.
	Book Retreat & Pay Deposit	M2	Guest selects package/dates/villa type and pays deposit via VNPay.
	Check-In & Assign Villa	M2	Receptionist verifies Guest, collects required residence information, and assigns a physical villa.
	Manage Villa Status	M2	Receptionist manages Available/Occupied/Maintenance operational status.
	View Booking & Itinerary	M2	Guest views booking details and itinerary timeline.
	Create Walk-in Booking	M2	The Receptionist checks villa availability and creates a retreat package/booking for a walk-in guest (a guest who has not booked in advance via the website).
	Schedule Spa/Therapy Session	M3 Spa Scheduling	Allows a Customer or Receptionist acting on behalf of a Customer to initiate a Spa/Therapy scheduling request. HSWRMS validates eligibility, generates suitable scheduling options, obtains final confirmation, reserves the required resources, creates the session, and synchronizes the confirmed schedule.
	Validate Customer & Service Eligibility	M3	Verifies that the authenticated Customer has a valid retreat/stay context and is eligible to use the selected included or additional Spa/Therapy service before scheduling proceeds.
	Generate Recommended Slot Options	M3	Generates and ranks suitable scheduling options based on the requested treatment, preferred date/time constraints, treatment duration, therapist qualification, Treatment Room compatibility, and current resource availability.
	Auto-Match Qualified Therapist & Compatible Treatment Room	M3	Identifies valid scheduling combinations in which a qualified Therapist/Yoga Instructor and a compatible Treatment Room are both available for the same complete treatment interval.
	Confirm Recommended Schedule	M3	Allows the Customer or Receptionist to review and confirm one of the system-recommended scheduling options before the resources are finally committed.

ID	Use Case	Feature	Use Case Description
	Revalidate Selected Schedule	M3	Rechecks the selected time slot, Therapist, and Treatment Room immediately before reservation to prevent conflicts caused by availability changes occurring after the recommendation was generated.
	Atomically Reserve Therapist & Treatment Room	M3	Reserves the selected Therapist and Treatment Room as one atomic scheduling operation. The reservation is valid only when both resources can be committed for the same complete interval; a partial reservation is not accepted.
	Book Extra Spa/Therapy Service	M3	Allows a Receptionist to initiate an additional Spa/Therapy service for an eligible Customer. The extra service uses the same automated scheduling, matching, confirmation, and resource-reservation process as a normal Spa/Therapy session.
	Post Extra Spa Charge to Customer Folio	M3	Posts the charge for a confirmed additional Spa/Therapy service to the Customer Folio so that it can later be included in the consolidated invoice and checkout process.
	Manage Pre-Session Confirmation	M3	Manages the Customer confirmation lifecycle before a confirmed Spa/Therapy session, including reminder delivery, confirmation response, expiration, and the decision to keep the reserved slot or initiate recovery scheduling.
	Record Customer Confirmation Response	M3	Records whether the Customer explicitly confirms, declines, or fails to respond before the confirmation deadline. An expired confirmation is not treated as a No-Show; final attendance is recorded later by the Therapist/Yoga Instructor.
	Sync / Update Confirmed Session to Customer Calendar	M3	Sends confirmed or changed Spa/Therapy session information to the external Calendar & Notification Services so that the Customer's calendar event can be created, updated, or cancelled as required.
	Send Reminder & Confirmation Request	M3	Sends the scheduled pre-session notifications, including the informational reminder, primary confirmation request, final reminder when required, and upcoming-session reminder for confirmed sessions.
	Notify Customer of Schedule Change	M3	Notifies the affected Customer whenever a confirmed session's time, Therapist, Treatment Room, or other relevant scheduling information changes after recovery or approved rescheduling.
	Auto-Reschedule / Re-Match Session	M3	Automatically attempts to recover a session when confirmation is declined/expired, an assigned Therapist becomes unavailable, or another scheduling disruption occurs. The system prioritizes free same-day alternatives, alternative qualified resources, another suitable day, and finally a Customer slot swap when no free alternative exists.

ID	Use Case	Feature	Use Case Description
	Propose Customer Slot Swap	M3	As a final fallback, proposes exchanging confirmed time slots between affected Customers when no suitable free alternative can be found. A proposed swap does not modify any confirmed schedule until all required Customer approvals are received and the resources are revalidated.
	Collect Explicit Consent from All Affected Customers	M3	Obtains explicit acceptance from every Customer whose confirmed schedule would change as part of a proposed slot swap. The swap may proceed only when all affected Customers accept; otherwise the existing schedules remain unchanged.
	View Daily Schedule & Treatment Safety Health Profile	M3	Allows a Spa Therapist/Yoga Instructor to view assigned sessions for the selected working day and access the Customer's complete treatment-safety health information required to identify contraindications and safely deliver the assigned treatment. Unrelated administrative, residence, payment, and account-security data remain hidden.
	Record Session Attendance & Outcome	M3	Allows the assigned Therapist/Yoga Instructor to record whether the Customer attended the session and to finalize the session outcome as Completed or No-Show after the scheduled treatment period.
	Update Ingredient Availability	M4 Dietary F&B	Staff update the quantity, quality, and usage limits of raw materials in the system.
	Update Menu Availability	M4	After staff update ingredient information, the system automatically checks the status of the ingredients and updates the related dishes to "Available" or "Unavailable" before customers make their selections.
	View Personalized Menu	M4	Customers can view the list of dishes available during their stay. The menu is filtered by allergy/diet profile, customer can only select items that are currently available.
	Pre-select Daily Meals	M4	Customer selects meals from a menu filtered by allergy/diet profile and meal type from the available options..
	Order Extra A-la-carte Item	M4	Customers can select and order additional items from the available menu and charges it to the customer folio, beyond those included in the Daily Meal or retreat package.
	View Meal Plan / Order Status	M4	Customers view their selected dishes, ordered a-la-carte items, and processing status.
	View Meal Preparation Dashboard	M4	Chef/F&B staff view the list of meal orders requiring preparation, including meal type, dish, quantity, and status.
	View Necessary Allergy/Dietary Alerts	M4	Staff view allergy or dietary alerts necessary for food preparation, not the full health profile.
	Update Meal Order Status	M4	Chef/F&B updates meal state from Preparing -> Ready->Delivered.

ID	Use Case	Feature	Use Case Description
	Minimize Health Data for Chef	M4	Only food-allergy/diet information required by F&B is exposed; unrelated medical history is hidden.
	Generate Consolidated Invoice	M5 Checkout & Analytics	Receptionist aggregates package, spa and F&B charges by Room_Booking_ID.
	Process Final Payment & Update Villa	M5	Receptionist completes payment only after outstanding charges are cleared, then updates villa status.
	Submit Review & Rating	M5	Guest submits a review after checkout.
	View Revenue Dashboard	M5	Manager views revenue breakdown by Package, Spa and F&B.
	Export Occupancy & Therapist Utilization	M5	Manager exports occupancy and therapist utilization reports to Excel.
	Configure Business Parameters	M6 System Administration	Admin configures parameters such as refund percentage and cancellation window.
	Toggle Feature Flags	M6	Admin enables/disables features by module.
	Configure Permission Matrix	M6	Admin configures detailed permission rules by screen/role.
	Delegate Limited Admin Rights	M6	Admin grants Manager a limited subset of administrative capabilities.
	View Sensitive-Data Audit Log	M6	Admin reviews logged access to sensitive information.
	View System Overview Dashboard	M6	Admin/Manager views a system-level operational overview according to granted access.

4.3 Use Case Diagrams

At the end of Week 2, the team maintains one use-case diagram per module and one consolidated system-wide use-case view. Detailed scenarios, preconditions, alternate flows, activity diagrams and screen flows are deferred to later iterations.

4.3.1 Module 1 - Authentication & Sensitive Health Profile: (Sai)

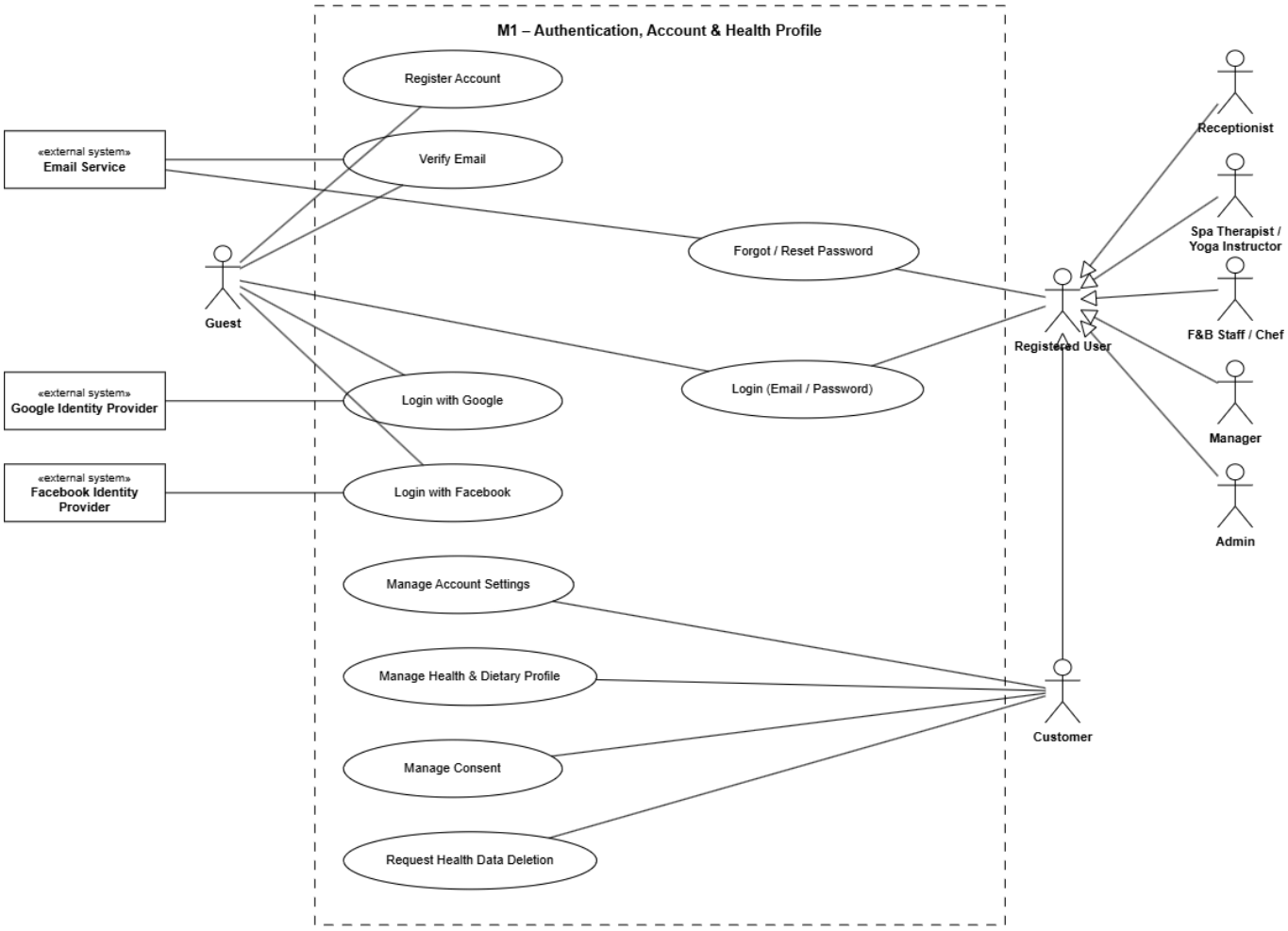
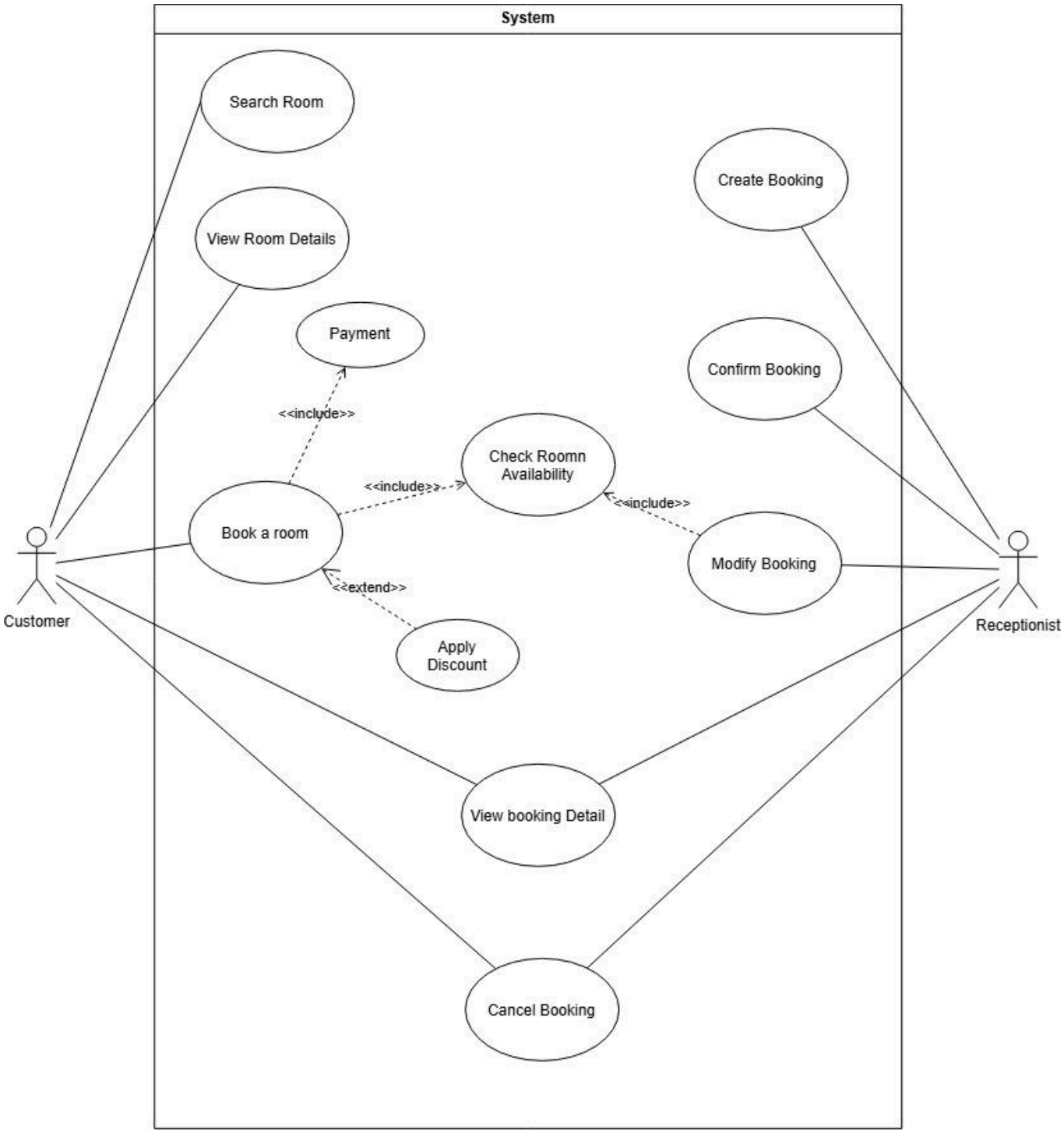


Figure 2. Module 1 Authentication, Account & Health Profile

Link Module 1 - Authentication,Account & Health Profile

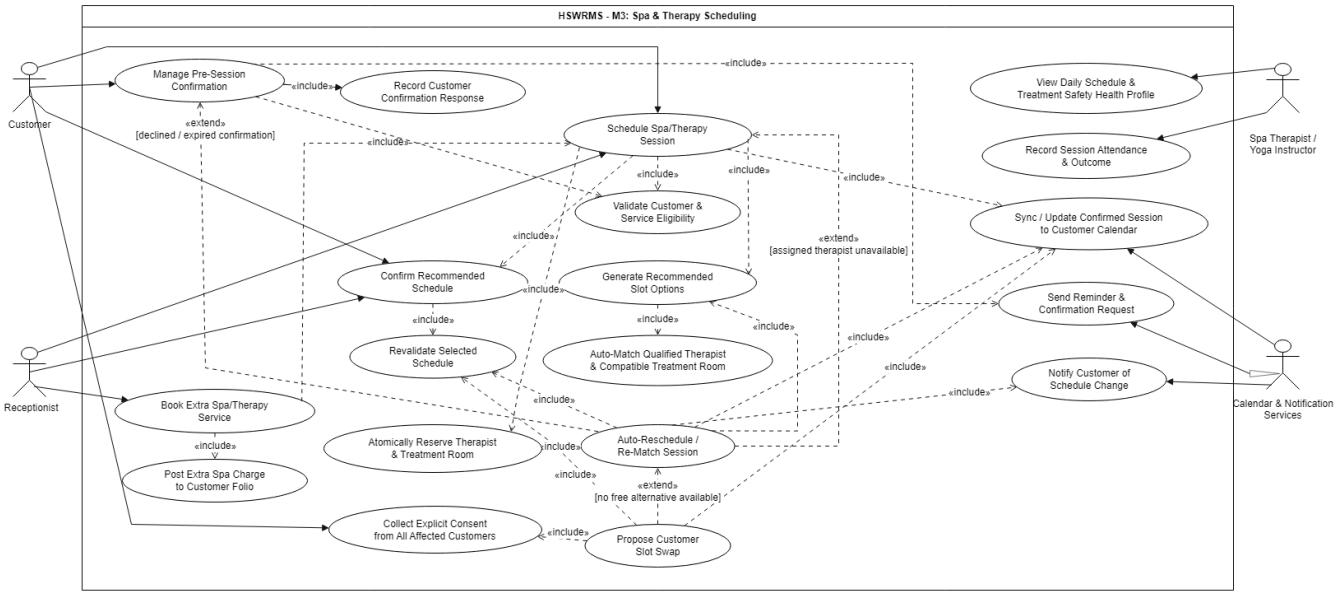
4.3.2 Module 2 - Retreat Package & Accommodation Booking:



Link Module 2 - Retreat Package & Accommodation Booking

Figure 3. Module 2 Retreat Package & Accommodation Booking

4.3.3 Module 3 - Spa & Therapy Scheduling:



[Link Module 3 - Spa & Therapy Scheduling]

Figure 4. Module 3 Spa & Therapy Scheduling:

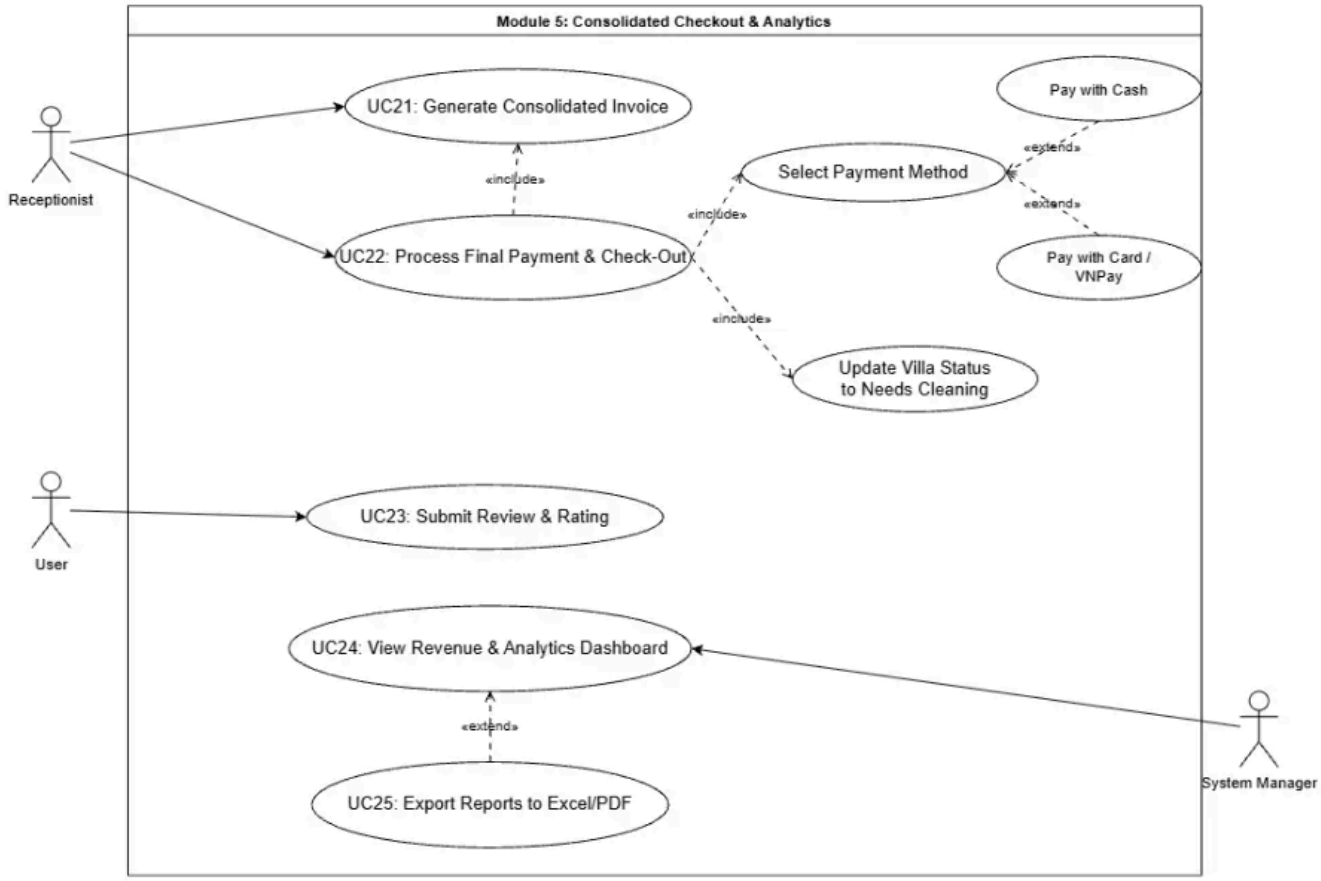
4.3.4 Module 4 - Dietary F&B Management:



Link Module 4-Dietary F&B Management

Figure 5. Module 4 Dietary F&B Management

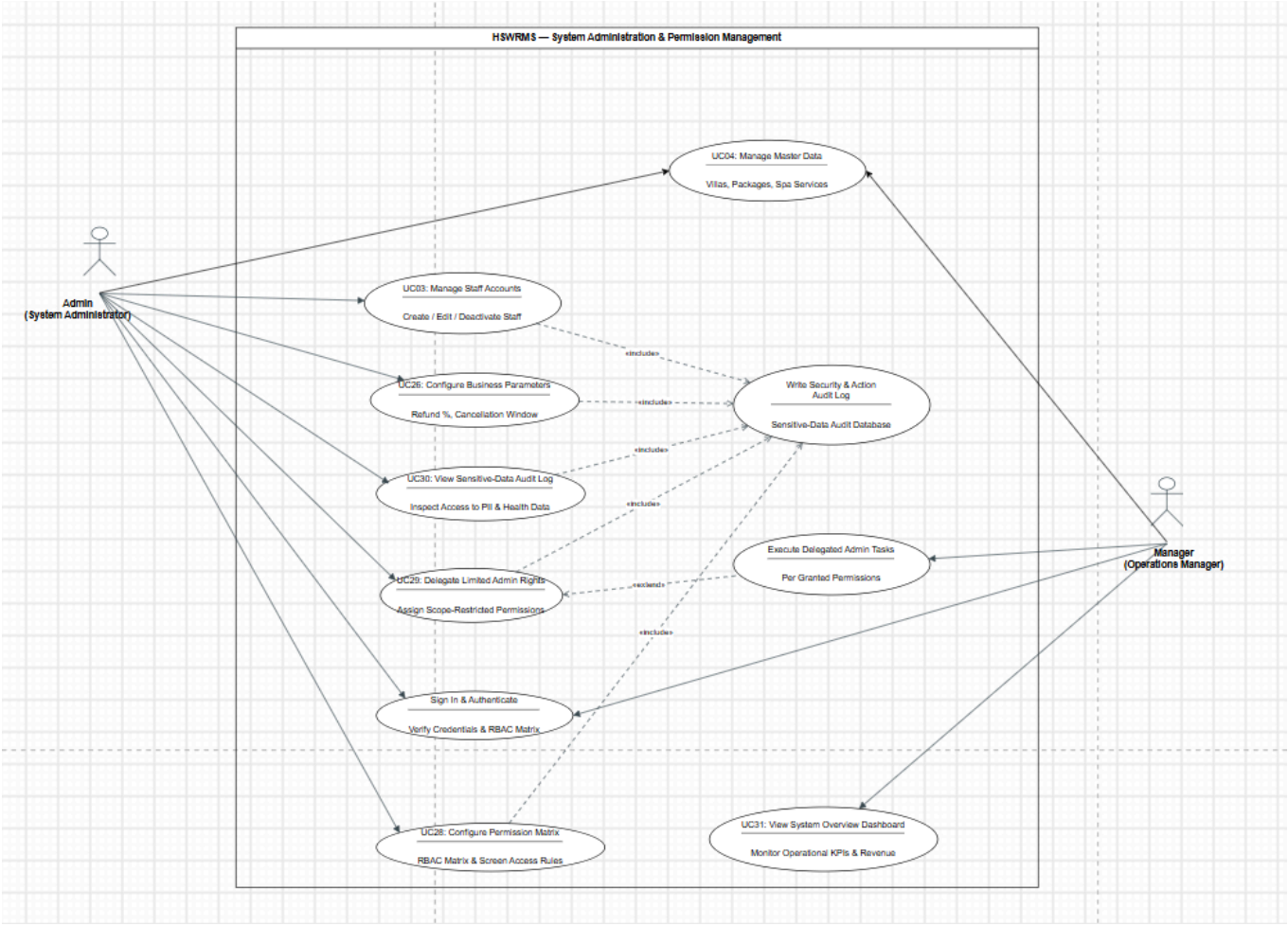
4.3.5 Module 5 - Consolidated Checkout & Analysis:



Link Module 5-Consolidated Checkout & Analysis

Figure 6. Module 5 Consolidated Checkout & Analysis

4.3.6 Module 6 - System Administration & Permission Management:



Link Module 6 - System Administration & Permission Management

Figure 7. Module 6 System Administration & Permission Management

4.3.7 Consolidated System-Wide Use Case View

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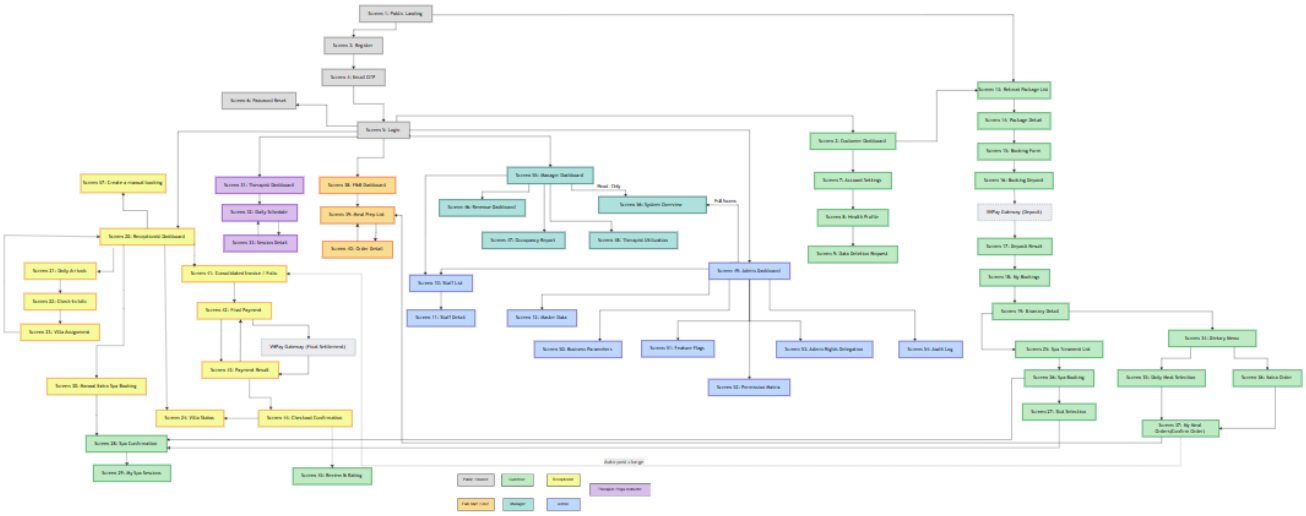
Figure 8. Consolidated System-Wide Use Case View

4.4 Permission Matrix

5. System Functionalities

5.1 Screen Inventory

a. Screen Flow



[Link Screenflow Draft]

b. Screen List

#	Screen Name	Feature	Description
1	Public Landing	Common / M1-M2	Public entry screen providing access to retreat packages, registration and login.
2	Customer Dashboard	Common / Customer Shell [Derived]	Main post-login navigation screen for Customer, providing access to bookings, spa sessions, meals, profile and folio.
3	Register	M1 – Authentication	Allows a new Customer to create an HSWRMS account using required personal and authentication information.
4	Email OTP Verification	M1 – Authentication	Allows Customer to enter the OTP sent by email to verify the registered account.

5	Login	M1 – Authentication	Allows users to sign in using email/password or initiate supported Google/Facebook SSO.
6	Password Reset	M1 – Authentication [Derived]	Allows a user to request and set a new password when login credentials are forgotten.
7	Account Settings	M1 – Authentication [Derived]	Allows an authenticated Customer to manage personal information, contact preferences, and account security settings, including password change.
8	Health & Dietary Profile	M1 – Health Profile	Allows Customer to view, create and update allergy, dietary, and treatment-relevant health information and manage related consent preferences.
9	Health Data Deletion Request	M1 – Health Profile	Allows Customer to request deletion or anonymization of eligible health and allergy data after retreat completion
10	Staff Account List	M6 – System Administration	Allows Admin to view and search staff accounts and their assigned roles/status.
11	Staff Account Detail	M6 – System Administration	Allows Admin to create, edit, assign roles to or deactivate a staff account.
12	Master Data Management	M6 – System Administration	Allows Admin to manage master data such as Villa Categories, Spa Services and Retreat Packages.
13	Retreat Package List	M2 – Retreat Booking	Displays available retreat packages and allows Customers to filter packages by wellness goal or relevant criteria.
14	Retreat Package Detail	M2 – Retreat Booking	Displays package description, included services, duration, villa options and information required before booking.
15	Retreat Booking Form	M2 – Retreat Booking	Allows customers to select retreat dates, package options and villa type and provide required booking information.

16	Booking Summary & Deposit	M2 – Retreat Booking	Shows booking summary, deposit amount and payment information before redirecting the Customer to VNPay.
17	Deposit Payment Result	M2 – Retreat Booking [Derived]	Displays the returned deposit payment result and confirmed booking information after VNPay processing.
18	My Bookings	M2 – Booking Management	Displays the Customer's current and historical retreat bookings.
19	Booking Detail & Itinerary	M2 – Booking Management	Displays booking status, retreat information, accommodation details and itinerary timeline.
20	Receptionist Dashboard	M2 – Front Desk [Derived]	Main operational navigation screen for Receptionist, including arrivals, check-in, villa operations and checkout access.
21	Daily Arrivals	M2 – Check-In	Displays Customers expected to arrive and allows Receptionist to open the relevant booking for check-in.
22	Check-In & Residence Information	M2 – Check-In	Allows Receptionists to verify the Customer and collect required identification and residence information for check-in.
23	Villa Assignment	M2 – Check-In	Allows Receptionist to assign an available physical villa to a confirmed booking during check-in.
24	Villa Status Management	M2 – Accommodation Operations	Allows Receptionists to view and update operational villa status such as Available, Occupied or Maintenance.
25	Spa Treatment List	M3 – Spa Scheduling	Displays Spa/Therapy services available to the Customer, including services included in the retreat package.
26	Spa Session Booking	M3 – Spa Scheduling	Allows Customer to select a Spa/Therapy service, date and requested time slot. The system validates both Therapist and Treatment Room availability.

27	Available / Alternative Slot Selection	M3 – Spa Scheduling	Displays available or alternative time slots when the originally requested Therapist and Treatment Room combination is unavailable.
28	Spa Booking Confirmation	M3 – Spa Scheduling	Displays confirmed Spa/Therapy session information after both Therapist and Treatment Room have been successfully reserved.
29	My Spa Sessions	M3 – Spa Scheduling [Derived]	Displays upcoming and historical Spa/Therapy sessions belonging to the Customer.
30	Manual Extra Spa Booking	M3 – Extra Spa Service	Allows Receptionists to select a Customer and manually book an extra Spa/Therapy service, with the charge posted to the Customer Folio.
31	Therapist Dashboard	M3 – Therapist Operations [Derived]	Main screen for Spa Therapist/Yoga Instructor showing assigned work and access to the daily schedule.
32	Therapist Daily Schedule	M3 – Therapist Operations	Displays Spa/Therapy sessions assigned to the logged-in Therapist for the selected working day.
33	Therapist Session Detail	M3 – Therapist Operations	Displays assigned session details and only treatment-relevant health notes; allows Therapist to mark the session Completed or No-Show.
34	Dietary Menu	M4 – Dietary F&B	Displays meal options filtered according to the Customer's relevant allergy and dietary profile.
35	Daily Meal Selection	M4 – Dietary F&B	Allows customers to pre-select meals for the applicable retreat day.
36	Extra A-la-carte Order	M4 – Extra F&B	Allows Customers to select and order additional F&B items that are charged to the Customer Folio.
37	My Meal Plan / Orders	M4 – Dietary F&B [Derived]	Allows customers to view selected daily meals and submit additional F&B orders.

38	F&B Dashboard	M4 – F&B Operations [Derived]	Main operational screen for Chef/F&B Staff providing access to daily meal preparation requirements.
39	Daily Meal Prep List	M4 – F&B Operations	Displays aggregated meal orders and preparation workload for Chef/F&B Staff.
40	Meal Order Detail	M4 – F&B Operations	Displays meal order details and only relevant allergy/dietary warnings; allows F&B Staff to update status from Preparing to Ready.
41	Customer Folio & Consolidated Invoice	M5 – Checkout 1	Displays consolidated Package, Spa and F&B charges for the selected stay and identifies any outstanding amount.
42	Final Payment	M5 – Checkout 2	Allows final payment to be initiated only when applicable charges have been consolidated and checkout requirements are satisfied.
43	Final Payment Result	M5 – Checkout [Derived]	Displays the result returned from the payment process and indicates whether payment was successfully confirmed.
44	Checkout Confirmation	M5 – Checkout	Displays successful checkout after payment is confirmed and the booking and villa operational statuses have been updated.
45	Review & Rating	M5 – Customer Feedback	Allows customers to submit a rating and review for a completed retreat stay, either from HSWRMS or an emailed review link.
46	Revenue Dashboard	M5 – Analytics	Allows Managers to view revenue information and breakdowns across Retreat Package, Spa and F&B services.
47	Occupancy Report	M5 – Analytics	Allows Manager to view accommodation occupancy information and export permitted report data.
48	Therapist Utilization Report	M5 – Analytics	Allows Manager to view Therapist utilization information and export permitted report data.

49	Admin Dashboard	M6 – System Administration [Derived]	Main administration navigation screen for Admin, providing access to configuration, permissions, security and audit functions.
50	Business Parameters	M6 – System Administration	Allows Admin to configure system business parameters such as cancellation windows, refund percentages and other configurable rules.
51	Feature Flag Management	M6 – System Administration	Allows Admin to enable or disable supported system features or modules using feature flags.
52	Permission Matrix	M6 – Permission Management	Allows Admin to configure detailed role-to-screen and role-to-function access permissions.
53	Limited Admin Rights Delegation	M6 – Permission Management	Allows Admin to grant or revoke a limited subset of administrative capabilities for Manager.
54	Sensitive Data Audit Log	M6 – Audit & Security	Allows Admin to review logged access and sensitive-data related activities for security and accountability.
55	Manager Dashboard	M6 – Management Monitoring [Derived]	Main navigation screen for Manager, providing access to authorized management dashboards, reports and system overview.
56	System Overview Dashboard	M6 – System Monitoring	Displays a system-level operational overview to Admin or Manager according to granted permissions.

5.2 External API Inventory

5.3 Background Job Inventory

II. Functional Requirements

1. Module 1 - Authentication & Sensitive Health Profile

1.1 UC01 - Register, Verify Email & Login

1.2 UC02 - Manage Dietary & Health Profile

1.3 UC03 - Manage Staff Accounts & Roles

1.4 UC04 - Manage Master Data

1.5 UC05 - Request Health Data Deletion

2. Module 2 - Retreat Package & Accommodation Booking

2.1 UC06 - Browse Retreat Packages

2.2 UC07 - Book Retreat & Pay Deposit

2.3 UC08 - Check-In & Assign Villa

2.4 UC09 - Manage Villa Status

2.5 UC10 - View Booking & Itinerary

3. Module 3 - Spa & Therapy Scheduling

3.1 UC11 - Schedule Included Spa/Therapy Session

3.2 UC12 - Match Therapist & Treatment Room

3.3 UC13 - View Therapist Schedule & Health Notes

3.4 UC14 - Mark Session Completed / No-Show

3.5 UC15 - Manually Book Extra Spa Service

4. Module 4 - Dietary F&B Management

4.1 UC16 - Pre-select Daily Meals

4.2 UC17 - View Daily Meal Prep Dashboard

4.3 UC18 - Update Meal Order Status

4.4 UC19 - Order Extra A-la-carte Item

4.5 UC20 - Minimize Health Data for Chef

5. Module 5 - Consolidated Checkout & Analytics

5.1 UC21 - Generate Consolidated Invoice

5.2 UC22 - Process Final Payment & Update Villa

5.3 UC23 - Submit Review & Rating

5.4 UC24 - View Revenue Dashboard

5.5 UC25 - Export Occupancy & Therapist Utilization

6. Module 6 - System Administration & Permission Management

6.1 UC26 - Configure Business Parameters

6.2 UC27 - Toggle Feature Flags

6.3 UC28 - Configure Permission Matrix

6.4 UC29 - Delegate Limited Admin Rights

6.5 UC30 - View Sensitive-Data Audit Log

6.6 UC31 - View System Overview Dashboard

7. Background Job Requirements

8. External API Requirements

III. Requirement Appendix

1. Business Rules

GBR-ID	Rule Statement	Applies To	Rationale	Enforced By
GB-01				

2. System Messages

#	Message code	Message Type	Context	Content
1	MSG01			

3. Other Requirements

4. Open Items Requiring Team Confirmation

5. References and Source Basis